

4.4.3 Hazards and uses of radioactive emissions and of background radiation (physics only)

4.4.3.1 Background radiation

Content	Key opportunities for skills development
Background radiation is around us all of the time. It comes from: • natural sources such as rocks and cosmic rays from space • man-made sources such as the fallout from nuclear weapons testing and nuclear accidents. The level of background radiation and radiation dose may be affected by occupation and/or location.	
Radiation dose is measured in sieverts (Sv) 1000 millisieverts (mSv) = 1 sievert (Sv) Students will not need to recall the unit of radiation dose.	WS 4.4

4.4.3.2 Different half-lives of radioactive isotopes

Content	Key opportunities for skills development
Radioactive isotopes have a very wide range of half-life values. Students should be able to explain why the hazards associated with radioactive material differ according to the half-life involved.	MS 1b Students should be able to use data presented in standard form.

4.4.3.3 Uses of nuclear radiation

Content	Key opportunities for skills development
Nuclear radiations are used in medicine for the: • exploration of internal organs • control or destruction of unwanted tissue. Students should be able to:	
• describe and evaluate the uses of nuclear radiations for exploration of internal organs, and for control or destruction of unwanted tissue	WS 1.4
• evaluate the perceived risks of using nuclear radiations in relation to given data and consequences.	WS 1.5